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6

Filtering & Functions In SQL

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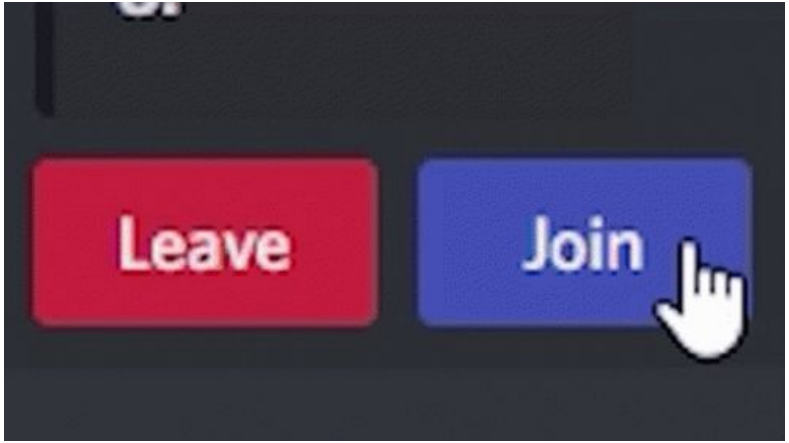

RECAP

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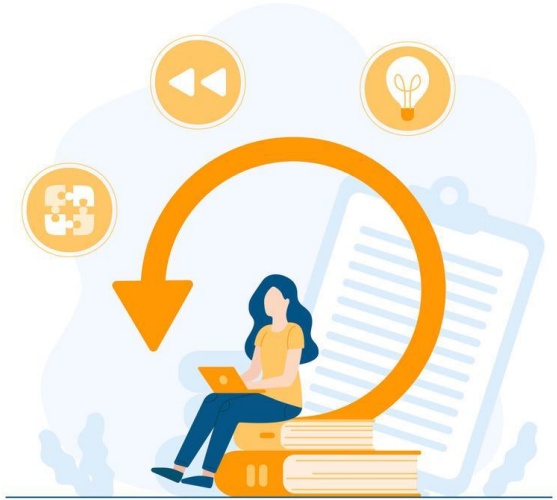
In last lecture, we learnt about:

- Order By Clause
- Limit Clause
- Distinct Clause
- Where Clause
- Comparison Operator
- Logical Operator
- Arithmetic Operators
- Other operators



Agenda

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Today, we will dive into: Operators and Functions

- **Wildcard Operators** - % and _
- **Aggregate Functions** - Count, AVG, SUM, MIN, MAX
- **Numeric Functions** - ROUND, CEIL, FLOOR, ABS, SQRT, ABS
- **String Manipulation Functions** - CONCAT, LEFT, RIGHT, INSTR, LOWER, TRIM, RTRIM, LTRIM, UPPER, ASCII, REPLACE, POSITION, SUBSTRING. & more
- **Null Handling Functions** - COALESCE
- **Nested Functions** - USING Multiple functions together like AVG and Ceil.

SQL File Link

Wildcards & Aggregates

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How do you search on Amazon/Flipkart?

By name, category, price range, or rating

What do you think is happening in the backend?

SQL is being used to **filter, search, and sort** data

How can retrieve rows based on some pattern?

Wildcards help us search for **patterns** in text (e.g.,names starting with "Sam")

Aggregate Functions help us:

Get **summary insights** like *total sales*, *average ratings*, etc.

Identifying Patterns

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How can we identify customers whose names starts with ‘R’?

Complaint_id	Customer_name	Severity
2	Mira	3
3	Rajat	2
1	Rahul	1
4	Amit	2
5	Mohit	3
6	Priya	3
7	Anmol	1

Wildcard Operators

Wildcard operators in SQL are some special characters that are used to perform pattern-matching searches in text data.

Syntax:

SELECT column1, column2, ...
FROM table_name
WHERE column_name **LIKE** pattern;

Wildcard Operators

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Some of the most commonly used wildcard operators are:

Operator	Description
%	Represents zero, one, or multiple characters.
_	Represents a single character.

Wildcard Operators

Query to display all the customer names starting with ‘R’ will be:

```
Select customer_name
from customer_complaints
Where customer_name like 'R%';
```

Identifying Patterns

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How can we retrieve customer_name whose name contains exactly 4 characters?

Complaint_id	Customer_name	Severity
2	Mira	3
3	Rajat	2
1	Rahul	1
4	Amit	2
5	Mohit	3
6	Priya	3
7	Anmol	1

Wildcard Operators

This query will retrieve all the customers whose name has exactly four characters will be:

SELECT * **FROM** customer_complaints
WHERE customer_name **LIKE** '_ _ _ _';

Write a query to retrieve all employee names from the employees table where the name contains the letter **'a'** followed by **any one character**, and then the letter **'e'** — anywhere in the name.

SELECT name
FROM employees
WHERE name LIKE '%a_e%';

Question Time

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1. What is the main use case of Wildcards

2. What does % and _ do in SQL wildcards?

3. What happens when you use LIKE '____' in SQL? What do the underscores represent?

4. How would you search for all users in your database whose last name ends with 'son'?

Revenue Generation

As business owner we want to see the total revenue generated, how can we calculate the sum of order amount?

Aggregate Functions

Aggregate functions in SQL perform calculations on a set of values and return a single value. These are essential for data analysis and reporting.

Syntax:

SELECT AGGREGATE_FUNCTION(column_name)

FROM table_name;

Aggregate Functions

Name	Type
COUNT	Count of the number of rows selected
AVG	Average of the values in a column
SUM	Total sum of values in a numeric column
MIN	Smallest Value in a column
MAX	Largest value in a column

Aggregate Functions

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Query to calculate the sum of order amount is:

```
Select sum(total_order_amount)
From Orders;
```

Write an SQL query to count the total number of orders whose status is ‘Delivered’.

SELECT COUNT(*) FROM Orders
WHERE status = 'Delivered';

Question Time

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1. What is the average price of all the products?

2. How would you calculate the highest order_amount and lowest order_amount of the orders?

3. What will the query “SELECT SUM(price) FROM Products;” return?

Counting Customer Names

How can we count customer names including NULL values?

Aggregate Functions

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Query to count customer names including NULL values will be:

SELECT

Count(*)

FROM customers;

Rounding To Nearest Integer

How can i round of the average order amount to their nearest lower integer or higher integer?

Numeric Functions

We can achieve the result using the query:

```
SELECT  Ceil(AVG(total_order_amount)
FROM orders;
```

What will be the output of the query
SELECT ROUND(123.4567, 2);

Output: 123.46

Numeric Functions

Numeric functions in SQL are essential tools that are used to perform mathematical operations, manipulation, and transformation of numeric data within database.

These functions are essential for calculations, data analysis, and transformations.

Numeric Functions

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Various type of Numeric Functions involve:

Operator	Description
ROUND()	Rounds the value to certain specified decimal points.
CEIL()	Rounds the value up to nearest integer.
FLOOR()	Rounds the value down to nearest integer.
SQRT()	Calculates Square root of a value.
ABS()	Returns the absolute value

Email Campaign

How can we change customers name to display in capital letters as we are planning for a promotional email campaign of NewtonMart ?

String Manipulation Function

To list customer names in capital letters we can write:

Select UPPER(name)

from Customers;

String Manipulation Function

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String Functions in SQL are built-in functions that allow users to manipulate character data in various ways.

String Manipulation Function

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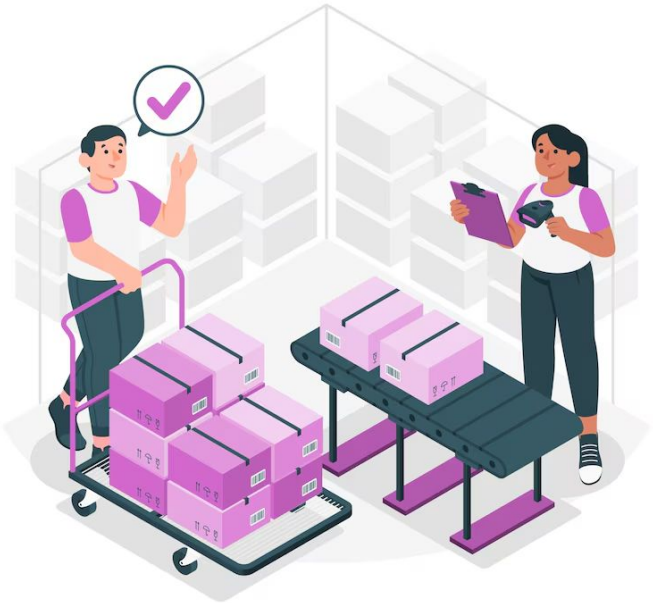
Sometimes the product name can be too long and we want only the first 5 characters of product names. How can we do that?



String Manipulation Function

The query to extract first five characters of product names would be:

Select product_id, SUBSTR(name, 1, 5)

From products;

String Manipulation Function

How can we replace product_name “Mixer Grinder” to “Grinder”?

String Manipulation Function

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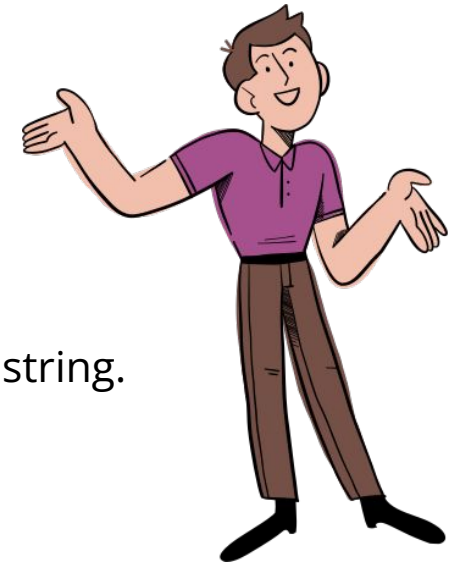
Query to replace product_name “Mixer Grinder” to “Grinder will be:

```
Select product_id,
REPLACE(name,”Mixer Grinder”,”Mixer “) As p_name
From products;
```



Replace helps to remove or replace data selectively from a string.

Syntax

```
REPLACE(String,CharToReplace,NewChar)
```

String Manipulation Function

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Most common string manipulation functions are listed below:

- **CONCAT** : Select Concat(name,description) as p_concat from products;

  It concatenates two or more strings together.

- **LEFT** : Select Left(name,3) as l_string from products;

  Extracts a number of characters from a string (starting from left)

- **RIGHT** : Select Right(name,3) as l_string from products;

  Extracts a number of characters from a string (starting from right)

- **INSTR:** Select INSTR(main_string, substring) from products;

  Returns the position of the first occurrence of a substring within a string.

String Manipulation Function

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- **LOWER** : Select lower(name) as p_lower from products;

  Converts a string to lower-case.

- **TRIM** : Select trim(name) as p_trim from products;

  Removes leading and trailing spaces

- **RTRIM** : Select Rtrim(name) as p_rtrim from products;

  Removes trailing spaces from a string

- **LTRIM** : Select Ltrim(name) as p_ltrim from products;

  Removes leading spaces from a string

String Manipulation Function

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- **ASCII** : Select Ascii(name) as p_ascii from products;

  Returns the ASCII value for the specific character

- **REPLACE** : Select Replace('SQL Tutorial', 'T', 'M');

  Replaces all occurrences of a substring within a string

- **POSITION** : Select Position("a" In "name");

  Returns the position of the first occurrence of a substring in a string

- **CAST** : Converts one data type into another, Useful when you want to apply string functions on numbers like phone numbers.

- **SUBSTRING** : Select Substring(name,1,3) as p_extractstring from products;

  Extracts some characters from a string

Question Time

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1. If a customer accidentally enters a name with extra spaces, how would you remove those spaces from the database?

2. What will the query SELECT CONCAT(first_name, ' ', last_name) FROM Customers; return?

3. Write a query to convert the first name of customers to uppercase and remove any leading/trailing spaces.

4. Write a query to display last 4 digits of phone number .

Replacing NULL Values

There are some Null values in the tables.How do i replace all of them?

IFNULL Function

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Query to replace Null value of product_description can be:

SELECT IFNULL (Product_description,”No info”) from products;

IFNULL takes two arguments. If the first argument is NULL, it returns the second argument; otherwise, it returns the first argument.

Syntax: IFNULL(expression, replacement_value)

Replacing NULL Values

In IFNULL we have the limitation of using only 2 expression,is there any other way to handle NULL values?

Coalesce Function

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Yes, the query to replace Null value of product_description with multiple expression will be:

SELECT name, COALESCE(description, 'No Description') AS description_filled FROM products;

SELECT name, COALESCE(NULLIF(TRIM(description), ''), 'No Description') AS description_filled FROM products;

Syntax:

COALESCE (expression1, expression2..., expressionx)

Coalesce Function

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This function evaluates the expression in a definite order and always results first not null value from the defined expression list.

Properties of the Syntax of SQL Server Coalesce function:

- All expressions must have the same data type.
- It could have multiple expressions.

Question Time

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1. What is the purpose of COALESCE() in SQL?
2. Write a query to replace NULL email addresses with 'Not Provided'.

Nested Functions

Let’s learn to apply functions within other functions.

The query will be:

```
SELECT ROUND(SUM(price), 2) FROM Products;
```

Nested Functions

How can we retrieve the total length of Customers name if we have two columns with first name and last name?

Combining length() and Concat()

The query will be:

```
SELECT LENGTH(CONCAT(first_name, ' ', last_name)) FROM Employees;
```

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Summary

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So far we have learnt

- **Wildcard Operator**
  Used in LIKE queries to search patterns within string values (e.g., %, _)

- **Aggregate Function**
  Functions like SUM(), AVG(), COUNT() etc. that summarize data over multiple rows

- **Numeric Function**
  Performs mathematical operations like ROUND(), CEIL(), FLOOR(), MOD(), etc.

- **String Manipulation Function**
  Helps to modify or extract parts of strings (e.g., CONCAT(), SUBSTRING(), LENGTH())

- **COALESCE Function**
  Returns the first non-null value from a list of expressions, useful for handling NULLs

Please fill the feedback form.

Additional References

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1. Extra questions to practice on Workbench
2. W3Schools SQL Tutorial - https://www.w3schools.com/sql/sql_aggregate_functions.asp
3. Leetcode - https://leetcode.com/problemset/database/
4. TutorialsPoint SQL Commands- https://www.tutorialspoint.com/sql/sql-aggregate-functions.htm
5. SQLZoo – Hands-on practice - https://sqlzoo.net/

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Thanks for watching!